

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

ILLUMINA, INC.,
Petitioner,

v.

MOLECULAR LOOP BIOSCIENCES, INC.,
Patent Owner.

IPR2024-01419
Patent 11,840,730 B1

Before ANNETTE R. REIMERS, SUSAN L. C. MITCHELL, and
ZHENYU YANG, *Administrative Patent Judges*.

MITCHELL, *Administrative Patent Judge*.

JUDGMENT

Final Written Decision
Determining No Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

A. BACKGROUND AND SUMMARY

Illumina, Inc. (“Petitioner”) filed a Petition (Paper 1, “Petition” or “Pet.”) seeking *inter partes* review of claims 1–9 of U.S. Patent No. 11,840,730 B1 (Ex. 1001, “the ’730 patent”). Molecular Loop Biosciences, Inc. (“Patent Owner”) filed a Preliminary Response to the Petition. Paper 6 (“Preliminary Response” or “Prelim. Resp.”). On April 25, 2025, we granted institution of an *inter partes* review of claims 1–9 of the ’730 patent on all grounds set forth in the Petition. Paper 10 (“Institution Decision” or “Dec.”).

During trial, Patent Owner filed a Patent Owner Response, (Paper 18 (“Response” or “PO Resp.”)), Petitioner filed a Reply (Paper 25 (“Reply”)), and Patent Owner filed a Sur-Reply (Paper 27 (“Sur-Reply”)). We held an oral hearing on January 29, 2026, and a transcript of that hearing is of record. Paper 33 (“Tr.”).

We have jurisdiction under 35 U.S.C. § 6(b). This is a Final Written Decision under 35 U.S.C. § 318(a) as to the patentability of claims 1–9. After considering the full record developed through trial, we determine as set forth below that Petitioner has not shown by a preponderance of the evidence that any of challenged claims 1–9 are unpatentable. *See* 35 U.S.C. § 316(e).

B. REAL PARTIES IN INTEREST

Petitioner identifies Illumina, Inc., as its real party-in-interest. Pet. 10. Patent Owner identifies Molecular Loop Biosciences, Inc. as its real party-in-interest. Paper 3, 1.

C. RELATED MATTERS

The parties indicate that the '730 patent is involved in *Molecular Loop Biosciences, Inc. v. Illumina, Inc. et al.*, No. 1:24-cv-00680 (D. Del.), filed June 10, 2024, which the parties indicate is currently pending. Pet. 10–11; Paper 3, 1. The parties also report that the '730 Patent was previously involved in *Illumina, Inc. et al. v. Molecular Loop Biosciences, Inc.*, No. 1:24-cv-11501 (D. Mass.). Paper 8, 1; Paper 9, 1–2.

D. THE '730 PATENT (EX. 1001)

The '730 patent is titled “Methods and Compositions for Evaluating Genetic Markers.” Ex. 1001, code (54). The '730 patent is directed to methods and compositions for determining genotypes in patient samples. *Id.* at 1:13–14.

The '730 patent discusses solving the problems of systematic and random error that are associated with genome amplification and sequencing library construction techniques. *See, e.g.*, Ex. 1001, 27:45–30:47. Specifically, the '730 patent states:

In certain embodiments, the disclosure provides preparative methods for capturing target nucleic acids (e.g., genetic loci) that involve the use of differentiator tag sequences to uniquely tag individual nucleic acid molecules. In some embodiments, the differentiator tag sequence permits the detection of bias based on the frequency with which pairs of differentiator tag and target sequences are observed in a sequencing reaction. In other embodiments, the methods reduce errors caused by bias, or the risk of bias, associated with the capture, amplification and sequencing of genetic loci, e.g. for diagnostic purposes.

Id. at 28:1–24.

These unique differentiator tag sequences are described in the '730 patent as

useful to distinguish independent target molecules from each other thereby allowing an analysis to be based on a known number of individual target molecules. For example, if each of a plurality of target molecule sequences obtained in an assay is associated with a different differentiator tag, then the target sequences can be considered to be independent of each other and a genotype likelihood can be determined based on this information. In contrast, if each of the plurality of target molecule sequences obtained in the assay is associated with the same differentiator tag, then they probably all originated from the same target molecule due to overrepresentation (e.g., due to biased amplification) of this target molecule in the assay.

Ex. 1001, 28:29–42.

The '730 patent further describes these unique differentiator tag sequences as follows:

As used herein, a “differentiator tag sequence” is a sequence of a nucleic acid (a preparative nucleic acid), which in the context of a plurality of different isolated nucleic acids, identifies a unique, independently isolated nucleic acid. Typically, differentiator tag sequences are used to identify the origin of a target nucleic acid at one or more stages of a nucleic acid preparative method. For example, in the context of a multiplex nucleic acid capture reaction, differentiator tag sequences provide a basis for differentiating between multiple independent, target nucleic acid capture events. Also, in the context of a multiplex nucleic acid amplification reaction, differentiator tag sequences provide a basis for differentiating between multiple independent, primary amplicons of a target nucleic acid, for example. Thus, combinations of target nucleic acid and differentiator tag sequence (target:differentiator tag sequences) of an isolated nucleic acid of a preparative methods provide a basis for identifying unique, independently isolated target nucleic acids.

Ex. 1001, 28:53–29:4.

E. THE CHALLENGED CLAIMS

Petitioner challenges claims 1–9 of the '730 patent. Pet. 1. Claims 1 and 9 are independent. Claim 1 is representative and is reproduced below:

1. A method for correcting for errors or bias introduced during nucleic acid analysis workflow, the method comprising the steps of:

obtaining a biological sample comprising a plurality of target nucleic acid molecules from more than one locus of origin;

introducing a set of differentiator tags, wherein members of said set of differentiator tags are associated with members of said plurality, such that one or more of said loci of origin are associated with more than one differentiator tag;

amplifying each of the plurality of tagged target nucleic acid molecules to generate amplicons;

sequencing the amplicons, obtained in said amplifying step to obtain sequence reads of each of the amplicons, wherein each of the sequence reads comprises a target nucleic acid molecule sequence and a differentiator tag sequence; and

correcting for error or bias introduced during said workflow by collapsing target:differentiator tag combinations observed more than once into a single count.

Ex. 1001, 131:28–49.

Claim 9 claims a method for correcting for errors or bias introduced during nucleic acid analysis workflow that is similar to claim 1, but explicitly requires “unique” differentiator tags that “are randomly associated with members of said plurality of target nucleic acid molecules, such that any given tag is uniquely associated with said locus of origin.” Ex. 1001, 132:44–48.

The correcting for error or bias step of claim 9 is also different from claim 1. Claim 9 requires that the correction is accomplished by

“recognizing target/differentiator tags having the same sequence as being derived from the same input molecule.” Ex. 1001, 132, 56–59.

F. EVIDENCE

Petitioner relies upon information that includes the following:

Ex. 1010, Chee, US 9,085,798 B2, issued July 21, 2015 (“Chee”).

Ex. 1015, Gahan et al., *Circulating Nucleic Acids in Plasma and Serum*, ANN. N.Y. ACAD. SCI. 1137:1–6 (2008) (“Gahan”).

Ex. 1013, Bignell et al., US 8,053,192 B2, issued Nov. 8, 2011 (“Bignell”).

Ex. 1012, McCloskey et al., US 2007/0020640 A1, published Jan. 25, 2007 (“McCloskey”).

Ex. 1017, Bentley, *Whole-genome re-sequencing*, Current Opinion in 16 GENETICS & DEVELOPMENT 545–52 (2006) (“Bentley”).

Ex. 1033, Chiang et al., *High-resolution mapping of copy-number alterations with massively parallel sequencing*, 6 NATURE METHODS 99–103 (2009) (“Chiang”).

Petitioner also relies on Declarations of Stacey Gabriel, Ph.D. (Ex. 1005; Ex. 1065) to support its contentions.

Patent Owner relies on Declarations of Gregory Porreca, Ph.D. (Ex. 2004); Patrick R.H. Waller, Ph.D. (Ex. 2005); and Joseph G. Puglisi, Ph.D. (Ex. 2023) to support its contentions.

G. ASSERTED GROUNDS OF UNPATENTABILITY

Petitioner asserts that claims 1–9 would have been unpatentable on the following grounds:

Ground	Claim(s) Challenged	35 U.S.C. § ¹	Reference(s)/Basis
1	1–4, 9	102	Chee
2	1–4, 7, 9	103(a)	Chee
3	5, 6, 8	103(a)	Chee, Gahan
4	1, 4–6, 9	103(a)	Bignell, McCloskey
5	2–3	103(a)	Bignell, McCloskey, Bentley
6	6, 8	103(a)	Bignell, McCloskey, Gahan
7	7	103(a)	Bignell, McCloskey, Chiang

H. CLAIM CONSTRUCTION

The challenged claims should be read in light of the Specification, as it would be interpreted by one of ordinary skill in the art. *In re Suitco Surface, Inc.*, 603 F.3d 1255, 1260 (Fed. Cir. 2010). Thus, we generally give claim terms their ordinary and customary meaning. *See In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007) (“The ordinary and customary meaning is the meaning that the term would have to a person of ordinary skill in the art in question.” (internal quotation marks omitted)); *see also* 37 C.F.R. § 42.100(b) (stating that claims are construed in IPRs according to the same standard as used in federal court).

Petitioner provides express constructions for three claim phrases: “set of differentiator tags,” “set of unique differentiator tags,” “locus of origin,” and “consensus base call.” *See* Pet. 15–20. Patent Owner does not dispute directly Petitioner’s proposed constructions. *See generally* PO Resp. By virtue of its arguments against Petitioner’s grounds of unpatentability, however, Patent Owner provides its interpretation of “collapsing

¹ Because the ’730 patent has an earliest possible effective filing date before March 16, 2013, it is not subject to the post-grant review provisions of the America Invents Act. *See* The Leahy-Smith America Invents Act, Pub. L. No. 112-29, 125 Stat. 284 (“AIA”) § (3)(n)(1)(A) (2011).

target:differentiator tag combinations observed more than once into a single count” as recited in claim 1 (the “collapsing step”) and “determining a consensus base call for the target nucleic acid molecule sequences” as recited in claim 2 (the “determining step”). PO Resp. 42–45.

We address the construction of these claim limitations below and address the “consensus base call” claim term in the context of the “determining step” of claim 2. *Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) (“[W]e need only construe terms ‘that are in controversy, and only to the extent necessary to resolve the controversy.’” (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999))).

1. “Set of [Unique] Differentiator Tags”

Petitioner asserts that both “set of differentiator tags” as used in claim 1 and “set of unique differentiator tags” as used in claim 9 should be given the same construction—“set of prepared tags where each tag includes a unique sequence as compared to the other tags.” *See* Pet. 16 (citing Ex. 1005 ¶ 89; Ex. 1001, 28:25–29, 28:53–57). Petitioner further asserts that “[t]hese unique differentiator tags are associated with target nucleic acids so that each nucleic acid is uniquely identifiable.” Pet. 16. Petitioner also points to the prosecution history of the ’730 patent in which Petitioner asserts the Patent Owner argued and the Board acknowledged that each differentiator tag is unique. *Id.* at 16–17. Petitioner concludes that “[a]n interpretation of ‘differentiator tags’ to include non-unique tags is not possible, and if it were, Applicant disclaimed any interpretation that would allow non-unique tags.” *Id.* at 17.

Patent Owner does not dispute Petitioner’s construction. *See generally* PO Resp.

On the full record, we agree with Petitioner that both claim phrases, “set of differentiator tags” and “set of unique differentiator tags” should be construed to mean “set of prepared tags where each tag includes a unique sequence as compared to other tags.” Petitioner has provided sufficient support for such a construction in the Specification of the ’730 patent as set forth below.

For instance, Petitioner points us to an express definition that links uniqueness with “differentiator tag,” which states:

[A] “differentiator tag sequence” is a sequence of a nucleic acid (preparative nucleic acid), which in the context of a plurality of different isolated nucleic acids, identifies a *unique*, independently isolated nucleic acid.

Ex. 1001, 28:53–57 (emphasis added). Petitioner also points us to the statement that “[a]spects of the invention relate to associating *unique* sequence tags (referred to as *differentiator tag* sequences) with individual target molecules that are independently captured and/or analyzed (e.g., prior to amplification or other process that may introduce bias).” Ex. 1001, 28:25–29 (emphasis added).

2. “*Locus of Origin*”

Both independent claims 1 and 9 use the term “locus of origin.” See Ex. 1001, 131:28–49, 132:38–59. Petitioner asserts that this term should be construed to mean “the location in a genome where a nucleic acid came from.” Pet. 17 (citing Ex. 1005 ¶ 96). Petitioner explains that Patent Owner added this term to clarify “that each molecule in the starting sample was associated with a different unique differentiator tag, even if multiple molecules were from the same genetic locus of origin.” Pet. 18–19 (citing Ex. 1005 ¶¶ 100–01).

Patent Owner does not dispute Petitioner's construction. *See generally* PO Resp.

Dr. Gabriel testifies that “[i]n the field of genetics, a ‘locus’ refers to a fixed location on a chromosome which contains a gene or other genetic marker.” Ex. 1005 ¶ 96. Dr. Gabriel further testifies that “[a] POSA would have understood that the ‘locus of origin’ of a nucleic acid refers to the location on a chromosome which contains the sequence that encodes for a given nucleic acid molecule.” *Id.* ¶ 98.

Dr. Gabriel points to statements in the Specification of the '730 patent that are consistent with her construction of “locus of origin.” *See* Ex. 1005 ¶¶ 96–97. For instance, the '730 Specification “describes a genetic loci as a location on the genome that can correspond to heritable disorder,” and “provides examples of genetic loci in Table 1.” *Id.* (citing Ex. 1001, 3:2–20, 3:43–47, 66:3–11, 31:50–42:12 (Table 1)). We also note that the '730 patent defines a “target nucleic acid” as a genetic locus. *See* Ex. 1001, 24:2–3.

On the full record, we credit Dr. Gabriel's testimony and determine that “locus of origin” refers to the location in a genome where a nucleic acid came from.

3. *The “Collapsing” and “Determining” Steps*

Patent Owner appears to accept Petitioner's construction of “consensus base call” as “using multiple sequencing reads of the same molecule to determine the nucleotide (base) at each position of a sequence,” PO Resp. 42 (citing Pet. 19), but states that “even under that interpretation, as the '730 Patent explains, ‘collapsing’ is directed toward fixing quantitative errors and is a different step distinct from determining a ‘consensus base call,’ which is directed toward fixing sequence errors,” *id.* (citing Ex. 2024, 46:10–25).

The “collapsing step” of claim 1 recites “correcting for error or bias introduced during said workflow by *collapsing target:differentiator tag combinations observed more than once into a single count.*” Ex. 1001, 131:46–48 (emphasis added). The “determining step” of claim 2 recites “determining a consensus base call for the target nucleic acid molecule sequences,” but performs “collapsing all sequence reads comprising a same differentiator tag into a single read,” *before* the “determining step.” *Id.* at 131:49–53. Patent Owner proposes an interpretation of these “collapsing” and “determining” steps recited in claims 1 and 2 and asserts that they should be construed as distinct steps performed in a particular order in which the “collapsing step” is performed before the “determining step.” PO Resp. 42–45. We agree.

First, Patent Owner asserts a person of ordinary skill in the art would have understood that the “collapsing step” of claim 1 is distinct from, and must occur before, the “determining step” of claim 2 based on the plain language of claims 1 and 2. PO Resp. 42–43. Patent Owner argues that the “collapsing step” and the “determining step” are distinct steps because claims 1 and 2 use different terms to describe these distinct steps that are performed sequentially according to the claim language. *Id.* (citing Ex. 1001, 131:47–54; Ex. 2023 ¶ 133; Ex. 2024, 57:11–58:22). More specifically, Patent Owner explains that claim 1 recites a “collapsing step,” followed by claim 2, which depends from claim 1 and refers to “collapsing” before “determining a consensus base call” such that “claim 2 describes determining a consensus base call from *collapsed* reads, *not* performing ‘collapsing’ by determining a consensus base call.” *Id.* at 42 (citing Ex. 1001, 131:47–54).

Patent Owner argues the specification of the '730 patent “explains that collapsing corrects for skewing, an issue related to over- or under-amplification, which is a separate and distinct problem from the sequence errors that a consensus base call is designed to address.” *Id.* at 43 (citing Ex. 1001, 27:47–50, 32:29–34). Patent Owner contends further that “[c]ollapsing” is a different step, performed separately from determining any consensus base call, to provide even greater accuracy when subsequently determining a consensus base call” such that “if one were to collapse by determining the consensus base call, errors from amplification bias could still be represented in the consensus base call generated.” *Id.* at 43–44 (emphasis omitted, citing Ex. 2023 ¶¶ 134–37).

Second, Patent Owner argues a person of ordinary skill in the art would have understood that the “collapsing step” of claim 1 is distinct from the “determining step” based on the specification of the '730 patent. PO Resp. 43–45. For example, Patent Owner explains that the specification states that “[f]or any target:differentiator tag combination observed more than once, all such reads are ‘collapsed’ into a single read before consensus basecalls are determined. This will cancel the effect of bias on consensus basecall accuracy.” *Id.* at 43 (emphasis omitted) (quoting Ex. 1001, 72:63–73:5) (citing Ex. 2023 ¶¶ 133, 138–142).

Moreover, Dr. Puglisi testifies that

a consensus base call is performed on already-collapsed sequence reads. And for good reason – by performing the consensus base call on collapsed sequence reads to determine a sample’s genotype, amplification bias will be removed from consideration. Without collapsing, amplification bias errors (i.e., quantitative errors) would still be considered in the consensus base sequence.

Ex. 2023 ¶ 133 (emphasis omitted). Dr. Puglisi further explains that “[b]efore determining a sample’s genotype at a given locus by performing a consensus base call, the method of the ’730 patent uses the ‘collapsing’ step to ensure that any errors in amplification bias do not carry through and affect the subsequent analysis.” *Id.* ¶ 135 (citing Ex. 1001, 72:63–73:5).

According to Dr. Puglisi, “[i]f the molecules with identical tag:combination sequences were not collapsed into one sequencing read before further analysis, the ’730 patent explains that considering sequence reads for all copies of a given original molecule could result in designating that locus with the wrong genotype.” *Id.*; *see* Ex. 1001, Fig. 5 (showing when over- or under-amplified and redundant sequence reads are not removed, “the skewed population of amplicons will result in generating a consensus base sequence using *all amplicons* and thus causing an **incorrect genotype call**.”).

Petitioner does not dispute Patent Owner’s construction. *See generally* Reply.

On review of the full record, we agree with Patent Owner that the “collapsing” and “determining” steps set forth in claims 1 and 2, respectively, are distinct from each other, and the “collapsing step” occurs before the “determining step.” Patent Owner’s proposed construction is consistent with the claim language and specification, and we credit Dr. Puglisi’s testimony as consistent with the record evidence.

I. LEVEL OF ORDINARY SKILL IN THE ART

Petitioner proposes that a person of ordinary skill in the art (“POSA” or “POSITA”) at the time of the invention “would have had (i) a Ph.D. in molecular biology, genetics, bioinformatics, or a related field, and at least two years of experience in high-throughput sequencing technologies, or (ii) a

Master’s degree in one of the same fields with at least four years of experience in high-throughput sequencing technologies,” with acknowledged flexibility in the varying combinations of education and practical experience. Pet. 10; Ex. 1005 ¶ 36. Petitioner states that a POSA “also commonly worked as part of an interdisciplinary team.” *Id.*

Patent Owner disagrees with Petitioner’s proposal that a POSA must have commonly worked as part of an interdisciplinary team because Petitioner does not explain what is required for the interdisciplinary team or “why a Ph.D. or Master’s degree in the relevant fields combined with an appropriate level of experience in high-throughput sequencing technologies would not suffice.” PO Resp. 34. Petitioner does not respond to Patent Owner’s argument in the Reply. *See generally* Reply.

On the full record, we adopt and apply Petitioner’s proposed level of ordinary skill because it does not appear to be inconsistent with the level of skill reflected in the asserted prior art. We do not interpret Petitioner’s statement that a POSA “also commonly worked as part of an interdisciplinary team,” however, as imposing a firm requirement that a POSA worked as part of an interdisciplinary team. We agree with Patent Owner that the statement is too vague to provide any guidance on the composition of such an interdisciplinary team. Accordingly, we do not consider the level of ordinary skill in the art to include any requirement that a POSA worked as part of an interdisciplinary team.

II. ANALYSIS

A. Introduction

“In an *inter partes* review, the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable.”

Harmonic Inc. v. Avid Tech., Inc., 815 F.3d 1356, 1363 (Fed. Cir. 2016) (citing 35 U.S.C. § 312(a)(3) (requiring *inter partes* review petitions to identify “with particularity . . . the evidence that supports the grounds for the challenge to each claim”)). This burden of persuasion never shifts to the patent owner. *See Dynamic Drinkware, LLC v. Nat’l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015).

1. Prior Invention

“To antedate (or establish priority) of an invention, a [patent owner] must show either an earlier reduction to practice, or an earlier conception followed by a diligent reduction to practice.” *Purdue Pharma L.P. v. Boehringer Ingelheim GmbH*, 237 F.3d 1359, 1365 (Fed. Cir. 2001). “Conception is the formation, in the mind of the inventor, of a definite and permanent idea of the complete and operative invention, as it is thereafter to be applied in practice.” *Cooper v. Goldfarb*, 154 F.3d 1321, 1327 (Fed. Cir. 1998). “A reduction to practice can be either a constructive reduction to practice, which occurs when a patent application is filed, or an actual reduction to practice.” *Id.* In order to establish an actual reduction to practice, the Patent Owner must establish that: (1) The inventors constructed an embodiment or performed a process that met all the limitations of the claimed invention; and (2) The inventors determined that the invention would work for its intended purpose. *See id.*

If a patent owner has not shown actual reduction to practice prior to the “critical date” of a reference, the patent owner may nonetheless antedate the reference by establishing prior conception and reasonable diligence towards a reduction to practice. *Purdue Pharma*, 237 F.3d at 1365.

“Reasonable diligence must be shown throughout the entire critical period, which begins just prior to the competing reference’s effective date and ends

on the date of the invention’s reduction to practice.” *Arctic Cat Inc. v. GEP Power Prods., Inc.*, 919 F.3d 1320, 1331 (2019). However, the “diligence need not be perfectly continuous—only *reasonably* continuous.” *Id.*

As discussed above, in an *inter partes* review, 35 U.S.C. § 316(e) imposes the ultimate burden of persuasion to “prove unpatentability by a preponderance of the evidence” on the petitioner. This burden never shifts to the patent owner. *Dynamic Drinkware, LLC v. Nat’l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015). However, when the patent owner attempts to antedate the prior art, “[a] second and distinct burden, the burden of production” can shift between the petitioner and the patentee. *Id.* at 1379; *see In re Magnum Oil Tools Int’l, Ltd.*, 829 F.3d 1364, 1375–76 (Fed. Cir. 2016). Specifically, the patent owner “bears the burden of establishing that its claimed invention is entitled to an earlier priority date than an asserted prior art reference.” *Magnum Oil Tools*, 829 F.3d at 1375–76. Once the patent owner establishes it is entitled to an earlier priority date, the burden of production then shifts back to the petitioner “to convince the court that [the patent owner] is not entitled to the benefit” of the earlier priority date. *Dynamic Drinkware*, 800 F.3d at 1379 (citing *Tech. Licensing Corp. v. Videotek, Inc.*, 545 F.3d 1316, 1328 (Fed. Cir. 2008)).

2. Anticipation

To show anticipation under § 102, each and every claim element, arranged as in the claim, must be found in a single prior art reference. *Net MoneyIN, Inc. v. VeriSign, Inc.*, 545 F.3d 1359 (Fed. Cir. 2008). The prior art need not, however, use the same words as the claims in order to find anticipation. *In re Gleave*, 560 F.3d 1331, 1334 (Fed. Cir. 2009). It is also permissible to take into account not only the literal teachings of the prior art reference, but also the inferences an ordinarily skilled person would draw

from the reference. *Eli Lilly and Co. v. Los Angeles Biomedical Res. Inst. at Harbor-UCLA Med. Ctr.*, 849 F.3d 1073, 1074–75 (Fed. Cir. 2017); *In re Preda*, 401 F.2d 825, 826 (CCPA 1968). A reference may also anticipate a claim even if it does not expressly teach all the limitations arranged or combined as in the claim, “if a person of skill in the art, reading the reference, would ‘at once envisage’ the claimed arrangement or combination.” *Microsoft Corp. v. Biscotti, Inc.*, 878 F.3d 1052, 1068 (Fed. Cir. 2017) (quoting *Kennametal, Inc. v. Ingersoll Cutting Tool Co.*, 780 F.3d 1376, 1381 (Fed. Cir. 2015)).

3. Obviousness

A claim is unpatentable under 35 U.S.C. § 103 if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. *See KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) objective evidence of nonobviousness, if any. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

In analyzing the obviousness of a combination of prior art elements, it can be important to identify a reason that would have prompted one of skill in the art “to combine . . . known elements in the fashion claimed by the patent at issue.” *KSR*, 550 U.S. at 418. A precise teaching directed to the specific subject matter of a challenged claim is not necessary to establish obviousness. *Id.* Rather, “any need or problem known in the field of

endeavor at the time of invention and addressed by the patent can provide a reason for combining the elements in the manner claimed.” *Id.* at 420.

Accordingly, a party that petitions the Board for a determination of unpatentability based on obviousness must show that “a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so.” *In re Magnum Oil Tools Int’l, Ltd.*, 829 F.3d 1364, 1381 (Fed. Cir. 2016) (internal quotation marks omitted). “Both the suggestion and the expectation of success must be founded in the prior art, not in the applicant’s disclosure.” *In re Dow Chemical Co.*, 837 F.2d 469, 473 (Fed. Cir. 1988).

An obviousness analysis “need not seek out precise teachings directed to the specific subject matter of the challenged claim, for a court can take account of the inferences and creative steps that a person of ordinary skill in the art would employ.” *KSR*, 550 U.S. at 418; *see In re Translogic Tech, Inc.*, 504 F.3d 1249, 1259 (Fed. Cir. 2007). In *KSR*, the Supreme Court also stated that an invention may be found obvious if trying a course of conduct would have been obvious to a POSITA:

When there is a design need or market pressure to solve a problem and there are a finite number of identified, predictable solutions, a person of ordinary skill has good reason to pursue the known options within his or her technical grasp. If this leads to the anticipated success, it is likely the product not of innovation but of ordinary skill and common sense. In that instance the fact that a combination was obvious to try might show that it was obvious under § 103.

550 U.S. at 421. “*KSR* affirmed the logical inverse of this statement by stating that § 103 bars patentability unless ‘the improvement is more than the predictable use of prior art elements according to their established

functions.” *In re Kubin*, 561 F.3d 1351, 1359–60 (Fed. Cir. 2009) (citing *KSR*, 550 U.S. at 417).

We analyze the asserted grounds of unpatentability in accordance with these principles to determine whether Petitioner has met its burden to establish unpatentability by a preponderance of the evidence.

B. SUMMARY OF THE CITED PRIOR ART

1. Chee (Ex. 1010)

Chee was filed on April 29, 2010, published on March 15, 2012, and issued on July 21, 2015. Ex. 1010, codes (22), (45), (65). Chee “relates to the field of oligonucleotide selection and synthetic biology and nucleotide determination.” *Id.* at 1:11–12. Chee recognizes that constructing multi-kilobase scale DNA molecules is costly because “de-novo chemical synthesis of the DNA building blocks is quite error-prone.” *Id.* at 1:38–41. To solve this problem, Chee asserts that “[t]he present invention addresses this need by providing constructs, sets of constructs, and methods for ensuring accuracy of oligonucleotides having a desired sequence,” which Chee identifies as “regions of interest” or ROI. *Id.* at 1:55–58, 2:5–9.

Chee further describes the invention as providing:

A set of oligonucleotide constructs, where the constructs of the set comprise an ROI and an identifier unique to the individual construct. It is a fundamental element of the invention that the unique identifier is associated with a construct, not the ROI; thus, constructs with ROIs having the correct sequence can be differentiated from constructs having substantially the same ROI but containing errors.

Ex. 1010, 2:21–28, 9:9–12 (“The constructs of the invention have a unique identifier associated with the initial construct, thus giving the ability to differentiate between constructs having substantially the same ROI.”).

Chee further states:

The unique identifiers associated with a particular ROI confirm that the amplification molecules are related, and thus sequence differences between molecules with the same ROI can be identified as technical errors rather than real differences in the sequence (e.g., sequence differences from two copies of a similar sequence in a sample). Furthermore, because molecules that are identical by descent can be identified, a consensus sequence can be determined using data from multiple molecules, thereby achieving a much higher accuracy rate than most conventional high throughput methods.

Ex. 1010, 9:29–39.

Chee concludes that “[t]he use of the uniquely identifiable constructs of the invention to ensure the determination, selection and/or isolation of error-free regions of interest (ROIs) are correct is extremely effective, and allows specific sequences to be selected with high accuracy and high throughput.” Ex. 1010, 2:9–14.

2. *McCloskey (Ex. 1012)*

McCloskey describes methods for verifying the authenticity of a nucleic acid molecule and its sequence using a molecular barcode and a batch-stamp, as well as authenticating nucleic acid amplification products such as polymerase chain reaction (“PCR”) products. Ex. 1012, Abstract, ¶¶ 2–4. More specifically, McCloskey states:

In the practice of the present invention nucleic acid molecules (e.g., genomic DNA fragments) are labeled with distinct sequence tags prior to PCR amplification. These sequence tags authenticate a nucleic acid sequence with information, such as the date of the experiment, and the sample identity. Thus, the present invention permits identification of valid sequences, and distinguishes the valid sequences from contaminants and redundant sequences arising from template re-cloning. Contaminant sequences can be identified even when multiple control (no DNA) PCR samples are negative.

Barcoding permits, for example, quantification of the relative abundance of genomic methylation patterns or polymorphic sequences by correcting for skewing that can arise from PCR amplification or the cloning of the products.

Ex. 1012 ¶ 5.

3. *Bignell (Ex. 1013)*

Bignell describes an invention that “relates to methods for indexing samples during the sequencing of polynucleotide templates, resulting in the attachment of tags specific to the source of each nucleic acid sample such that after a sequencing run, both the source and sequence of each polynucleotide can be determined.” Ex. 1013, Abstract, 1:24–28. Bignell noted that it would be advantageous to genotype large populations in a small number of reactions, but be able to identify the source of the individual samples in those reactions to “yield the information content of a full non-pooled population screen in the time and at the cost of a pooled reaction.” *Id.* at 2:3–8. Bignell states that to accomplish this, “DNA from more than one source can be sequenced on an array if each DNA sample is first tagged to enable its identification after it has been sequenced.” *Id.* at 3:3–6. To this end, Bignell describes methods “for sequencing nucleic acid sequences on an array and identifying subsets of nucleic acid sequences on an array, wherein each subset of nucleic acid sequences is isolated from a different source.” *Id.* at 3:60–64.

4. *Gahan (Ex. 1015)*

Gahan describes methods of isolating and analyzing nucleic acids from blood. *See* Ex. 1015, 1. According to Gahan, “[t]he ability to isolate, quantify, and analyze these molecules has led to the identification of specific nucleic acid fragments related to particular disorders such as diabetes,

cancer, myocardial infarction, and stroke, th[e]reby permitting their early diagnosis.” *Id.* Moreover, Gahan reports that analysis of maternal plasma for maternal and fetal DNA has been used to determine fetal sex and for the early identification of “a range of maternal complications such as preeclampsia and intrauterine growth retardation, as well as in fetal disorders including Rhesus incompatibility and β -thalassemia.” *Id.* at 3–4.

5. *Bentley (Ex. 1017)*

Bentley describes several DNA sequencing techniques and discusses researchers’ efforts to improve them. *See generally* Ex. 1017. For example, Bentley explains that “[c]apillary sequencing using Sanger chemistry generates high-quality raw sequence data over most of the length of each read.” *Id.* at 549. Users analyzing the data may “choose a threshold for filtering the data depending on the application. For example, multiple reads from the same sample can be aligned and used to derive a consensus base-call at each position. In this case the depth of the consensus provides high confidence base-calling.” *Id.*

6. *Chiang (Ex. 1033)*

Chiang describes techniques for determining copy number alterations of genes using massively parallel sequencing and a computational algorithm named SegSeq. *See generally* Ex. 1033. Chiang explains that copy number alterations of genes are significant because “[c]ancer results from somatic alterations in key genes, including point mutations, copy-number alterations and structural rearrangements. A powerful way to discover cancer-causing genes is to identify genomic regions that show recurrent copy-number alterations (gains and losses) in tumor genomes.” *Id.* at 99.

C. THE CHEE GROUNDS (GROUND 1–3)

Petitioner asserts that claims 1–4 and 9 are anticipated by Chee (Ground 1); that claims 1–4, 7, and 9 would have been obvious over Chee (Ground 2); and that claims 5, 6, and 8 would have been obvious over Chee and Gahan (Ground 3). Pet. 3, 20–45. Claims 1 and 9 are independent, and claims 2–8 depend from claim 1. Ex. 1001, 131:49–132:36. Petitioner asserts that Chee teaches every element of independent claims 1 and 9. See Pet. 25–32, 35–38. Petitioner asserts the obvious ground over Chee to rebut any argument from Patent Owner that “Chee does not disclose the claimed ‘*differentiator tags*’ because Chee states that its unique identifiers ‘need not be absolutely unique.’” *Id.* at 38 (citing Ex. 1010, 9:60–63, 11:22–25). Petitioner posits that “[e]ven if that distinction were accepted, configuring Chee’s technique to use identifiers that were absolutely unique would have been obvious in view of Chee’s own teachings.” *Id.*

As to claim 7, which adds “the step of determining the copy number of genomic region or transcript in a patient from whom the sample is obtained” to claim 1, Petitioner asserts:

A POSA would have recognized that analyzing the DNA sequences generated from Chee’s method to determine a copy number would have been beneficial and important for detecting and diagnosing certain diseases. A POSA would have also known that copy number analysis may be influenced by bias and would have been motivated to use Chee’s methods to identify and correct for bias.

Pet. 42 (citing Ex. 1005 ¶¶ 212–14; Ex. 1033, 101).

As to Ground 3 involving claims 5, 6, and 8, Petitioner relies on Gahan as teaching isolating bacterial or viral nucleic acids (claim 5), tumor nucleic acid molecules (claim 6), and maternally-circulating fetal or

placental nucleic acid molecules (claim 8) from a biological sample. *Id.* at 44–45.

Patent Owner asserts that Petitioner has not proved by a preponderance of the evidence that Chee anticipates or renders obvious any challenged claim, with or without combination with Gahan, for the following reasons. Patent Owner asserts that: (1) Chee is not prior art to claims 1–3 and 9 of the '730 patent; (2) Chee does not disclose or teach the “collapsing step” recited in claim 1; (3) Chee’s provisional application does not disclose all of the elements Petitioner relies on in the Chee grounds; (4) Chee does not disclose or teach that the “collapsing step” occurs before the “determining step” as recited in claim 2; and (5) Petitioner does not show that a person of ordinary skill in the art would have combined Chee and Gahan. *See* PO Resp. 35–51.

1. Whether Chee is Prior Art to the Challenged Claims

Patent Owner argues that Chee is not prior art to claims 1–3 or 9 of the '730 patent because these claims are entitled to a priority date of at least April 29, 2009. Patent Owner contends the inventors conceived of the inventions set forth in these claims at least as early as March 6, 2009, and diligently reduced these inventions to practice during the critical period of April 29, 2009 through May 18, 2009. PO Resp. 21–34; Sur-Reply 1–13.

Petitioner argues Patent Owner has not shown prior conception of both a method for correcting errors and a method for correcting bias, nor has Patent Owner demonstrated reasonable diligence in constructively reducing the invention to practice during the critical period. Reply 20–27.

We discuss the parties’ arguments in greater detail below.

a) Conception

To establish prior conception, Patent Owner relies primarily upon declarations from Dr. Gregory Porreca (an inventor listed on the '730 patent) and Dr. Patrick R.H. Waller (one of the attorneys responsible for prosecuting the '730 patent), correspondence between Dr. Porreca and attorneys responsible for prosecuting the '730 patent, and drafts of a provisional patent application Dr. Porreca sent to Dr. Waller on March 6, 2009 (the “March 6 application,” Exs. 2006–2008). PO Resp. 23–30 (citing Exs. 2004–2008).

Patent Owner argues Dr. Porreca sent part of the March 6 application to the attorneys who prosecuted the '730 patent on March 6, 2009. PO Resp. 23–24 (citing Ex. 2004 ¶¶ 8–9; Ex. 2005 ¶ 7; Ex. 2006; Ex. 2007; Ex. 2008). Patent Owner provides claim charts indicating where the March 6 application discloses the claim limitations of claims 1–3 and 9. *Id.* at 24–30.

Petitioner does not dispute that the inventors of the '730 patent sent the draft application to the attorneys responsible for prosecuting the patent on March 6, 2009, or that the March 6 application demonstrates possession of all of the limitations of claims 1–3 and 9 except the “correcting for error bias” limitation. Reply 20–23. Petitioner asserts that the March 6 application “discloses at most a method for correcting bias” because the application does not include the word “error,” and Patent Owner’s witness, Dr. Joseph Puglisi, “admitted that sequencing errors are not disclosed in the March 6 Application or accompanying figures.” *Id.* at 21–22 (citing Ex. 1066, 57:22–59:1).

Petitioner contends Patent Owner “must show conception of both a method for correcting errors and a method for correcting bias” in order to demonstrate prior conception of claims 1–3 and 9 as evidenced by the March 6 application. Reply 21 (emphasis omitted). Petitioner argues error

must be different from bias because the terms would be superfluous of each other in the claim language if they mean the same thing. *Id.* at 22.

Petitioner asserts that “errors can occur in DNA synthesis during amplification or sequencing in which a nucleotide is incorrectly incorporated in the complementary strand,” whereas bias “occurs when molecules are over- or under-amplified in the amplification step.” *Id.* (citing Ex. 1005 ¶¶ 75–76; Ex. 1065 ¶¶ 38–43).

Patent Owner responds that portions of the March 6 application (Exhibit 2006) and Figure 2 of that application (Exhibit 2008) support correcting for error or bias. PO Resp. 24, 28–29 (quoting Ex. 2006, 1–3, 5; Ex. 2008). Patent Owner states that correcting for error and correcting for bias are overlapping concepts, so Petitioner is incorrect “that a wholly separate ‘method of correcting for errors’ needed to be described in addition to a ‘method of correcting’ for conception.” Sur-Reply 2–3. Patent Owner asserts that the claims cover one method that can correct for error or bias because the claims “merely recite a ‘method for correcting for errors or bias’ and a step of ‘correcting for errors or bias.’” *Id.* at 2. Patent Owner contends “[t]he steps in the claimed method, and the manner in which the correcting step is performed, do not differ based on whether ‘error’ or ‘bias’ is being corrected.” *Id.* In view of this overlap, Patent Owner contends “all evidence [Patent Owner] provided in the March 6 Application for ‘correcting for errors or bias’ applies equally to show conception of the claimed method of correcting for either.” *Id.* at 3.

Patent Owner asserts that Petitioner is also incorrect that the March 6 application does not include the word “error” because the application “discusses ‘error,’ and detecting such errors, including ‘synthesis errors.’” Sur-Reply 3, n.1 (citing Ex. 2006, 5). Patent Owner contends “the March 6

Application discloses using its methods to correct for overrepresentation, which is a type of error,” a concept to which both Dr. Puglisi and Dr. Gabriel agree. *Id.* at 4 (citing Ex. 1066, 11:19–25; Ex. 2023 ¶ 72; Ex. 2025, 27:20–28:5). For example, Patent Owner quotes a portion of the March 6 application stating “if we also have information on random barcodes, this situation could be averted, because the over-represented allele would be seen to also have an over-represented random barcode (i.e., the sequences with the overrepresented barcode all originated from the same single molecule).” *Id.* (emphasis omitted, citing Ex. 2006, 5).

Patent Owner contends Petitioner further errs by applying an unduly narrow interpretation of “error” that is not supported by intrinsic or extrinsic evidence. Sur-Reply 5–8. Patent Owner states that “[t]he plain language of the claims requires only ‘errors or bias’ that are ‘introduced during nucleic acid analysis workflow.’” *Id.* at 5. For example, “dependent claim language confirms that the ‘error’ recited in Claim 1 must be broader than only sequencing or amplification errors” because “[d]ependent Claim 3 recites an additional step of ‘determining the presence of a sequencing or amplification error based upon a number of said differentiator tag sequences.’” *Id.* at 5–6. Patent Owner argues that “the specification explains that systematic and random errors may themselves be bias, or bias may cause an error” because “the ’730 patent explains that its ‘methods [] overcom[e] problems associated with systematic and random errors (bias) in genome capture, amplification and sequencing method.’” *Id.* at 6 (citing Ex. 1001, 4:57–66, 5:8-11; Ex. 1068, 142:14–22, 143:11–19).

We have considered both parties’ arguments and evidence. We agree with Petitioner that Patent Owner must demonstrate that the March 6 application provides support for both correcting errors and correcting bias;

however, we agree with Patent Owner that the steps in the claimed methods do not differ based on whether error or bias is being corrected. The plain language of claims 1 and 9 requires support for both error and bias because the claims recite correcting for both. *See* Ex. 1001, 131:28–48, 132:39–60.

We determine Patent Owner has shown that the March 6 application demonstrates the inventors’ conception of “correcting for errors or bias.” Patent Owner provides several examples of correcting for bias in the March 6 application. PO Resp. 28–30. Patent Owner also explains that the March 6 application describes correcting for errors by describing using its methods to correct for overrepresentation. Sur-Reply 3–4; *see also* PO Resp. 28. More specifically, the March 6 application explains as follows.

[I]t is possible that a locus that is heterozygous is called as homozygous, because there are only a few observations of the second allele out of many observations of that locus.

However, if we also have information on random barcodes, this situation could be averted, because the over-represented allele would be seen to also have an overrepresented random barcode (i.e., the sequences with the overrepresented barcode all originated from the same single molecule). Therefore, the sequences and corresponding distribution of random barcodes can be used as an additional input to the genotype-calling algorithm to significantly improve the accuracy and confidence of the genotype calls.

Ex. 2006, 5 (emphasis added). In this example, the error occurs when the locus that is heterozygous is called as homozygous because of the error resulting from bias in the representation of the allele.

Based on this evidence of record, we determine that Patent Owner has shown that the March 6 application demonstrates conception of the invention recited in claims 1–3 and 9.

b) Reduction to Practice

To show reduction to practice, Patent Owner relies primarily upon declarations from Dr. Gregory Porreca (an inventor listed on the '730 patent) and Dr. Waller (one of the attorneys responsible for prosecuting the '730 patent), and correspondence between Dr. Porreca and attorneys responsible for prosecuting the '730 patent. PO Resp. 31–34 (citing Exs. 2004, 2005, 2012–2016). Patent Owner contends “Dr. Porreca and the prosecuting attorneys exercised reasonable diligence from at least April 29, 2009 through May 18, 2009” by drafting, editing, and filing Provisional Patent Application No. 61/179,358 (the “’358 application,” Ex. 1008). *Id.* at 31 (citing Exs. 1008, 2004, 2005, 2012–2016). Patent Owner asserts that “[b]y filing the ’358 provisional, Dr. Porreca and the inventors constructively reduced the ’730 patent to practice on May 18, 2009.” *Id.* at 32.

Patent Owner asserts the following activities demonstrate reasonable diligence.

- April 24, 2009: “Dr. Porreca received an initial set of claims for the application that became the ’730 patent.” PO Resp. 31 (citing Ex. 2004 ¶ 10; Ex. 2005 ¶ 10; Ex. 2012, 1).
- April 27, 2009: “Dr. Porreca provided his edits [to] the prosecuting attorneys.” *Id.* (citing Ex. 2004 ¶ 11; Ex. 2005 ¶ 11; Ex. 2012, 3).
- April 28, 2009: “Dr. Porreca received a revised set of claims for the application that became the ’730 patent.” *Id.* (citing Ex. 2004 ¶ 11; Ex. 2005 ¶ 11; Ex. 2012, 3).
- April 29, 2009: “Dr. Porreca spoke with the prosecuting attorneys regarding the inventions that became the ’730 patent.” *Id.* (citing Ex. 2004 ¶ 12; Ex. 2005 ¶ 12; Ex. 2012, 1–2).

- April 30, 2009: “Dr. Porreca provided further edits to the draft patent application for the prosecuting attorneys to add.” *Id.* at 32 (citing Ex. 2004 ¶¶ 12–13; Ex. 2005 ¶¶ 12–13; Ex. 2012, 1–2; Ex. 2013).
- May 5, 2009: “Dr. Porreca provided more materials to add to the application” and “met with the prosecuting attorneys to discuss the application that became the ’730 patent.” *Id.* (citing Ex. 2004 ¶¶ 14–15; Ex. 2005 ¶¶ 14–15; Ex. 2012, 1; Ex. 2014, 4–5).
- May 8, 2009 and May 9, 2009: “Dr. Porreca set up a call with the prosecuting attorneys that took place on May 9.” *Id.* (citing Ex. 2004 ¶ 16; Ex. 2005 ¶ 15; Ex. 2014, 3).
- May 11, 2009 and May 13, 2009: “Dr. Porreca exchanged emails with the prosecuting attorneys regarding the application that became the ’730 patent.” *Id.* (citing Ex. 2004 ¶¶ 17–18; Ex. 2005 ¶¶ 16–17; Ex. 2014, 1).
- May 14, 2009: “Dr. Porreca received an updated draft of the application that became the ’730 patent from the prosecuting attorneys.” *Id.* (citing Ex. 2004 ¶¶ 19–21; Ex. 2005 ¶ 18–20; Ex. 2015, 1; Ex. 2016).
- May 15, 2009: “Dr. Porreca provided his edits” to the application. *Id.* (citing Ex. 2004 ¶¶ 19–21; Ex. 2005 ¶ 18–20; Ex. 2015, 1; Ex. 2016).
- May 18, 2009: “[T]he prosecuting attorneys filed the ’358 provisional.” *Id.* (citing Ex. 1008; Ex. 2004 ¶ 22; Ex. 2005 ¶ 21).

Patent Owner also contends “Dr. Waller explains that he spent an average of 7 hours a day on his intellectual property cases, regularly

communicated with the inventor regarding the application, and created multiple drafts of the patent application throughout the entire critical period.” PO Resp. 33 (citing Ex. 2005 ¶¶ 1–21; Exs. 2013–2016).

Petitioner responds that Patent Owner fails to demonstrate reasonable diligence because Patent Owner “provides no evidence showing that any prosecution work was performed on 12 days of the 20-day critical period.” Reply 23. Petitioner argues “the evidence shows that Dr. Waller did not work on the ’358 Provisional, and instead delegated preparation of it to Dan Young, a patent agent at the time” such that “Dr. Waller’s testimony about his general practices or hours worked is not relevant to diligence on the ’358 Provisional because he did not draft it.” *Id.* at 24 (citing Ex. 1028; Ex. 1045; Ex. 1054; Ex. 1067, 57:17–58:21). Petitioner contends Patent Owner’s evidence is insufficient because Patent Owner did not submit a declaration from Dr. Young, billing records, or “evidence of what prosecution work, if any, Dr. Young did during the critical period.” *Id.* Moreover, Petitioner contends “[o]f the five days with activity, on two the only activity was a phone call, but there is no evidence of what was discussed or if the discussion involved the ’358 Application.” *Id.* at 26.

Patent Owner responds that Petitioner’s arguments fail because they (1) demand an unreasonably high requirement of constant seven-days-per-week work by ignoring that five of the twelve days are weekend days; (2) improperly ignore evidence on certain days or dismiss it as not showing enough work on that day; and (3) ignore evidence in communications on days surrounding the alleged inactive days that show that work was being done on the days between the communications.

Sur-Reply 9 (emphasis and footnote omitted).

Patent Owner provides additional information for each of the dates Petitioner alleges Patent Owner has not shown reasonable diligence. Sur-

Reply 10–11. Notably, Patent Owner explains that emails between the inventors and the prosecution attorneys show that the inventors sent paragraphs containing placeholders to insert into the application. *Id.* at 10. According to Patent Owner, those placeholders were “for ongoing modeling work for which ‘Uri is still calculating the actual numbers.’” *Id.* (quoting Ex. 2012, 1–2). Patent Owner contends the inventors followed up with another email that said “[w]e’re finished working out Uri’s [co-inventor] example’ and provided the results of the work,” which shows that the “work that went into the modeling/calculating results was done in the days in between (May 1-4) and evidence activity during those days.” *Id.* (citing Ex. 2012, 1). Patent Owner also points to “communications between the inventors and attorneys” that “mention the ongoing work for ‘wrapping up Uri’s application,’ including ‘completing the histogram for [the] application’” and setting a time to discuss the status of the work. *Id.* (quoting Ex. 2014, 4).

Regarding Patent Owner’s reliance on testimony from Dr. Waller rather than Dr. Young, Patent Owner states “Dr. Waller testified about his own diligent work throughout the critical period, including that he worked closely with Dr. Young on the application, did work on it himself, and did not merely hand it off to Dr. Young without oversight.” Sur-Reply 12 (citing Ex. 1067, 30:8–14, 30:19–31:4, 37:5–18, 38:17–22, 45:20–46:1, 46:14–47:13, 49:11–13, 57:7–15, 57:21–58:8). Patent Owner contends “[t]he documentary record reflects corroborating communications and drafts from Dr. Waller’s work on the application.” *Id.* (citing Ex. 2013, 1; Ex. 2014, 1–5; Ex. 2015, 1–2).

We have considered both parties’ arguments and evidence and determine that Patent Owner has presented sufficient evidence of reasonable

diligence. There is no requirement that an inventor must work on reducing the invention to practice every day during the critical period in order to demonstrate reasonable diligence. *See Perfect Surgical Techniques, Inc. v. Olympus America, Inc.*, 841 F.3d 1004, 1009 (Fed. Cir. 2016). “[D]iligence need not be perfectly continuous—only *reasonably* continuous.” *Arctic Cat Inc. v. GEP Power Prods., Inc.*, 919 F.3d 1320, 1331 (2019). The key question is whether, “in light of the evidence as a whole, the invention was not abandoned or unreasonably delayed.” *Id.* (internal quotations omitted). Applying this standard, we determine that Patent Owner sufficiently demonstrates reasonably continuous diligence throughout the critical period.

The evidence demonstrates that Patent Owner did not unreasonably delay filing the ’358 application. The evidence shows Dr. Porreca and the prosecuting attorneys worked diligently towards filing the ’358 application from April 24, 2009 through May 18, 2009 by drafting claims, editing the draft application, and performing additional modeling experiments to add to the specification. *See* Exs. 2004, 2005, 2012–2016. The prosecuting attorneys sent an updated draft of the ’358 application to Dr. Porreca on Friday, April 24, 2009. Ex. 2012, 4. On Monday of the following week, April 27, 2009, Dr. Porreca confirmed the draft looked good to him, and relayed a question from his co-inventor Dr. Uri Laserson. *Id.* at 3–4. The prosecuting attorneys sent revised claims to Dr. Porreca the following day, and planned to meet to discuss the claims on April 29, 2009. *Id.* at 2.

Dr. Porreca emailed the prosecuting attorneys on April 30, 2009 to provide the inventorship information (Ex. 2013) and to provide paragraphs to add to the draft application (Ex. 2012, 1–2). Dr. Porreca’s email says: “Note the two [XXXX] placeholders -- Uri is still calculating the actual numbers, but in the interest of time we decided to get this to you now, and will drop those

in when available,” which shows that Dr. Porreca and Dr. Laserson made efforts to avoid delays. *Id.* at 2 (emphasis omitted).

Dr. Porreca and Dr. Laserson finished the examples to complete the placeholders between April 30, 2009 and May 5, 2009. Ex. 2012, 1. Dr. Porreca emailed the prosecuting attorneys on May 5, 2009 and provided the paragraph the attorneys should use to fill in the placeholders. *Id.* The same day, the prosecuting attorneys followed up with some additional questions about the examples, and Dr. Porreca answered them. *Id.* The prosecuting attorneys asked for additional examples stating: “The second paragraph of the example below states that the specific number of barcodes necessary for a diploid basecall depends on the level of confidence desired and is calculated from the binomial probability distribution. Could you expand on this concept, perhaps by way of an example.” *Id.* Dr. Porreca replied that “We’re working on this and will have a histogram for inclusion hopefully tomorrow.” *Id.* The prosecuting attorneys provided another draft claim later that afternoon. Ex. 2014, 4–5. These email exchanges provide ample evidence that the inventors were working on examples for the application, and the prosecuting attorneys were continuing to draft claims between emails.

On Wednesday, May 6, 2009, Dr. Porreca emailed the prosecuting attorneys regarding “next steps” from a call that took place on May 5, 2009. Ex. 2014, 4. He stated that “you expect to be wrapping up Uri’s application,” and “Uri is completing the histogram for his application this morning.” *Id.* Dr. Porreca requested to schedule a call with the prosecuting attorneys for Friday, May 8, 2009. *Id.* at 3–4. They scheduled a call for Saturday, May 9, 2009. *Id.* at 2–3. On Monday, May 11, 2009, Dr. Porreca emailed the prosecuting attorneys to follow up on the May 9, 2009 call. *Id.*

at 2. He stated that “we should expect drafts on the two outstanding provisionals soon (and Uri’s probably doesn’t need further revision at this point). We’d like to give our investors a hard date for when we will send the provisionals for their review. Can you help me estimate this date?” *Id.* The prosecuting attorneys responded the same day and stated that “we owe you the two drafts and are working to get them to you as soon as possible. There should not be much left to finish, but I will talk to Dan and we will call you with the date.” *Id.* at 1.

Dr. Porreca sent a follow up email to the prosecuting attorneys on May 13, 2009 and asked, “Wanted to follow up on this -- do you have a sense for when we should expect either (or both) to be ready for final review and filing?” Ex. 2014, 1. The prosecuting attorneys responded the same day and stated that “Dan will forward Uri’s [application] tomorrow during the day.” *Id.* On May 14, 2009, the prosecuting attorneys provided an update to say that “Dan is back from exams this morning and is finalizing Uri’s application that he will send to you.” *Id.* Later that day, the prosecuting attorneys followed up with a second email with a draft of the application, and indicated that “We are still thinking about the term ‘source sequence’, and may change it,” and that they “included some specific questions in bold.” Ex. 2015, 1. Dr. Porreca provided revisions to the application on May 15, 2009, and addressed the prosecuting attorneys’ concerns. *Id.*; Ex. 2016, 1. Dr. Porreca stated that “We just got another investor inquiry to look at our IP under CDA, so we look forward to getting this one on file soon.” Ex. 2016. These emails also provide ample evidence that Dr. Porreca made efforts to avoid delays, and that Dr. Porreca and the prosecuting attorneys understood that time was of the essence to complete the application. The application was filed on May 18, 2009. Ex. 1008, 1.

We are not persuaded by Petitioner's arguments regarding Dr. Waller and Dr. Young's respective contributions to the '358 application because the email exchanges between Dr. Porreca and the prosecuting attorneys show that both Dr. Waller and Dr. Young were working diligently on the application and communicating with Dr. Porreca. *See, e.g.*, Exs. 2012–2016. Moreover, aside from pointing out that Dr. Waller understandably does not remember certain details about the work completed during the critical period that took place many years ago, Petitioner has not set forth any evidence that gives us any reason to doubt Dr. Waller's testimony.

Considering all of the pertinent evidence set forth above, we determine that Patent Owner did not abandon or unreasonably delay the filing of the '358 application during the critical period. Petitioner's attempts to scour the timeline for periods of alleged inactivity are unpersuasive. We therefore conclude that Patent Owner has established that it was reasonably continuous in its diligence during the critical period, and Petitioner has not shown otherwise.

c) Conclusion Regarding the Priority Date of the '730 Patent

Based on the foregoing, we determine Patent Owner has shown by a preponderance of the evidence that claims 1–3 and 9 are entitled to a priority date of at least April 29, 2009. As a result, Chee is not prior art to those claims of the '730 patent.

2. Ground 1 – Anticipation by Chee

After considering all of the arguments and cited evidence of record, we determine that Petitioner has not demonstrated by a preponderance of the evidence that Chee anticipates claims 1–4 and 9. As discussed above, Chee does not qualify as prior art to claims 1–3 and 9. Moreover, we determine

Chee does not disclose the “collapsing step” recited in claim 1, which is also required by dependent claim 4, for the reasons discussed below.

a) The “Collapsing Step” of Claim 1

Petitioner argues Chee discloses the “collapsing step” recited in claim 1. Pet. 31–32; Reply 11–15. First, Petitioner asserts that “Chee corrects for both errors and bias.” Pet. 31. More specifically, Petitioner contends “Chee discloses that ‘technical bias’ can be introduced when amplifying the original sample, which can ‘skew’ the final sample to have more copies of a particular construct as compared to others” such that “[b]y using the unique identifiers, ‘the quantification of the molecules can be corrected to reflect this manner of bias.’” *Id.* (citing Ex. 1010, 9:18–21, 10:36–46, 23:59–64; Ex. 1005 ¶¶ 172–74).

Second, Petitioner argues “Chee corrects bias by ‘*collapsing target:differentiator tag combinations . . . into a single count.*’” Pet. 31. Petitioner asserts that Chee’s unique identifiers allow it to “determine that two sequenced constructs are ‘identical by descent,’ *i.e.*, copies of the same construct from the original sample” and “accurately determine ‘the number of starting molecules . . . by association of each initial ROI present in a sample with its unique identifier.’” *Id.* at 31–32 (citing Ex. 1010, 5:27–36, 9:5–9, 10:36–46, 14:42–50, 24:1–8). Petitioner contends “Chee corrects for amplification bias by counting each unique construct once.” *Id.* (citing Ex. 1010, 23:59–64).

Third, Petitioner argues “Chee corrects for ‘sequencing errors’ by sequencing multiple copies of the same original molecule (those with the same unique identifier) and comparing the ROI reads.” Pet. 32 (citing Ex. 1010, 9:34–39, 10:54–57, 14:39–50). Petitioner asserts that Chee

discloses that “[m]ultiple reads of the same molecule are used to generate a single consensus read” because Chee states that:

[A] variation in the sequence of a single copy of a construct is most likely to be a synthesis or sequencing error if it differs from a consensus sequence derived from multiple clonal copies of the construct, which are identified as identical by descent via the shared unique identifier.

Id. (quoting Ex. 1010, 14:46–50; citing Ex. 1005 ¶¶ 176–78; Ex. 1017, 550).

Patent Owner responds that Chee does not disclose the “collapsing step” because “collapsing is different from the consensus base sequence in Chee’s method, and the quantitative errors corrected by collapsing are different from sequence errors corrected by determining a consensus base sequence.” PO Resp. 46 (citing Ex. 2024, 46:2–25). More specifically, Patent Owner asserts that Chee describes reducing sequence errors by “determining a consensus sequence through reading ‘multiple clonal copies of the construct, which are identified as identical by descent via the shared unique identifier,’ and attributing ‘a variation in the sequence of a single copy of a construct’ to an error occurring during sequencing or synthesis of the construct.” *Id.* (emphasis omitted, citing Ex. 1010, 14:39–56; Ex. 2023 ¶¶ 148–53; Ex. 2024, 12:17–13:1, 13:15–24, 14:42–50). Patent Owner contends this is different from the “collapsing step” recited in claim 1 as “[c]ollapsing does not determine a consensus base call by analyzing multiple reads of the same construct, because collapsing does the opposite; it assumes that all sequence reads for a single count will be identical, so there is no need to analyze multiple reads.” *Id.* (citing Ex. 2023 ¶¶ 158–63). Patent Owner argues that in this manner, “collapsing resolves quantification errors, allowing identical sequence reads to be counted only once without considering the precise sequence.” *Id.* (citing Ex. 2023 ¶¶ 154–56).

Moreover, Patent Owner argues Petitioner points to language from Chee that mentions using Chee's method to address issues related to amplification bias, but does not identify any mechanism to do so. PO Resp. 47. Patent Owner contends "beyond pointing to language that tags can be used to represent amplicons that are 'identical by descent,' Petitioner does not, and cannot, point to any precise method of Chee that explains how a POSA would use this information to count the number of original starting molecules of a given sequence." *Id.* Patent Owner also argues that while Chee states that "its method 'can' be used to identify amplification bias by associating each construct with a unique identifier," Petitioner "fails to cite to any disclosure to explain what that method entails." *Id.*

Petitioner responds that Chee teaches the "collapsing step" because Chee teaches processes that can correct for amplification bias and sequencing errors. Reply 11–15. According to Petitioner,

In Chee's process, after obtaining an initial sample, it adds a unique identifier to each nucleic acid in the sample to create constructs. EX1010(Chee), 4:1-3, 10:41-46. Each construct includes a unique identifier and a target nucleic acid (region of interest, "ROI"). *Id.*, 2:34-38. The sample is amplified (*e.g.*, using PCR), creating multiple copies of each construct, and the amplified constructs are sequenced. *Id.*, 3:10-17, 13:20-61. Chee explains that if multiple constructs have the same unique identifier, those constructs are "identical by descent," meaning they are copies of the same molecule from the original sample. *Id.*, 5:27-36, 14:42-46.

Reply 12.

Given this information, Petitioner contends Chee teaches correcting amplification bias because "its unique identifiers allow 'quantification of the molecules [to] be corrected to reflect [amplification] bias'" and "it can determine 'the number of starting molecules' in a sample by using the fact

that ‘each initial ROI present in a sample [was associated] with its unique identifier.’” Reply 12–13 (citing Ex. 1010, 5:31–36, 9:29–39, 10:41–46, 23:59–64; Ex. 1066, 120:1–5, 121:12–20). Petitioner argues Chee “recognizes that reads with the same unique identifier correspond to the same nucleic acid from the original sample and counts those once. . . . Thus, Chee’s process ‘collapses’ sequences with the same unique identifier into ‘a single count.’” *Id.* (emphasis omitted, citing Ex. 1010, 9:34–39, 10:41–46, 14:46–50, 23:59–64; Ex. 1005 ¶¶ 173–78).

Petitioner contends Chee teaches correcting sequencing errors “by comparing reads from copies of the same molecule and using them to create a ‘consensus’ sequence.” Reply 14 (citing Ex. 1010, 9:34–39, 10:54–57, 14:39–50; Ex. 1005 ¶¶ 177–78). Petitioner argues further that even if “collapsing” does not mean determining a consensus sequence, “Chee still anticipates or renders obvious claim 1” because “Chee’s technique for correcting amplification bias corrects quantitative errors without correcting sequencing errors or making consensus base calls.” *Id.* at 15.

Patent Owner responds that “[t]he claims require ‘correcting for error or bias introduced during said workflow **by collapsing target:differentiator tag combinations**.’” Sur-Reply 14. Patent Owner contends Petitioner fails to show Chee discloses the “collapsing step” because “Petitioner points to the same single step, determining a consensus base call from multiple clonal copies of a single molecule, to show both ‘collapsing’ and ‘determin[ing] a consensus base sequence.’” *Id.* at 15. Moreover, Patent Owner contends Chee’s method of correcting amplification bias does not disclose the “collapsing step” because “[e]ach of Chee’s disclosures simply conclude that Chee’s method quantifies molecules to correct for amplification bias. . . . But quantifying molecules to correct for amplification bias does not, without

more, disclose collapsing ‘target:differentiator tag combinations’ into a single count as required by the claims,” and Petitioner does not “point to any evidence to say that using a unique identifier to count each molecule discloses collapsing in the manner required by the Challenged Claims.” *Id.* (citing Ex. 1010, 10:41–46, 23:59–64; Ex. 1066, 120:1–5; Reply 13).

We have considered both parties’ arguments and evidence and agree with Patent Owner’s position. As discussed above, we agree with Patent Owner that the “collapsing step” recited in claim 1 and the “determining step” recited in claim 2 are two different steps required by the challenged claims such that “collapsing” does not mean determining a consensus base call or sequence, and must occur before any determining step. As a result, we determine Chee’s step of determining a consensus sequence does not disclose the “collapsing step” recited in claim 1 that requires collapsing “target:differentiator tag combinations.”

We are also unpersuaded by Petitioner’s reliance on Chee’s process of correcting for amplification bias as disclosing the claimed collapsing step. As recognized by Patent Owner, the claim language of the “collapsing step” requires “collapsing target:differentiator tag combinations,” rather than collapsing only differentiator tags. In view of this requirement, Chee does not disclose the “collapsing step” because Chee teaches quantifying the starting nucleic acid molecules using their unique identifiers, rather than target:differentiator tag combinations. As Petitioner admits, “Chee’s technique for correcting amplification bias corrects quantitative errors without correcting sequencing errors” (Reply 15), whereas claim 1’s “collapsing step” corrects both quantitative errors and sequencing errors by using target:differentiator tag combinations. *See* Ex. 1001, 131:47–49

(claim 1 requires correcting for error or bias by collapsing target:differentiator tag combinations).

As Dr. Puglisi confirms claim 2’s “collapsing step” corrects for both quantitative and sequencing errors by explaining “[t]he specification also describes ‘collapsing’ as resolving issues related to skewing while using the ‘collapsed’ sequences to derive a more accurate consensus base call for genotyping.” Ex. 2023 ¶ 134 (citing Ex. 1001, 65:2–14, 28:3–9). We credit Dr. Puglisi’s testimony as consistent with the teachings of the Specification of the ’730 patent.

Chee does not describe this claimed collapsing step using target:differentiator tag combinations to correct for both quantitative and sequencing errors. For example, Chee describes correcting using only differentiator tags in the following passage that mitigates only quantitative errors, not sequencing errors.

By creating constructs comprising the nucleic acids directly isolated from a sample, bias due to subsequent manipulation of the molecules can be identified **by determining the amounts of constructs with unique identifiers**. Thus, if a nucleic acid is either over-represented or under-represented due to amplification bias, the number of molecules comprising unique identifiers will also be over-represented or under-represented, and the quantification of the molecules can be corrected to reflect this manner of bias.

Ex. 1010, 23:56–64 (emphasis added).

For these reasons, we determine Petitioner has not shown that Chee discloses the “collapsing step” recited in claim 1.

b) Conclusion Regarding Anticipation by Chee

Based on the foregoing, we determine Petitioner has not shown by a preponderance of the evidence that Chee anticipates claims 1–4 and 9 of the

'730 patent. We determine Chee does not qualify as prior art to claims 1–3 and 9. We also determine Chee does not disclose claim 1's "collapsing step."

As a result, Chee does not anticipate claim 4, which depends from claim 1. Because Petitioner's anticipation ground fails for the above-described reasons, we need not address whether Chee's provisional application discloses all of the elements Petitioner relies on in the Chee grounds, or whether Chee discloses or teaches that the "collapsing step" occurs before the "determining step" as recited in claim 2. PO Resp. 35–41, 48–49.

3. Ground 2 – Obviousness over Chee

After considering all of the arguments and cited evidence of record, we determine that Petitioner has not demonstrated by a preponderance of the evidence that Chee renders claims 1–4, 7, and 9 obvious because Chee does not qualify as prior art to claims 1–3 and 9 for the reasons discussed above, and also as discussed above, Chee does not disclose the "collapsing step" recited in claim 1, which is a requirement of claims 2–4 and 7 by virtue of their dependence on claim 1. Pet. 38–39.

4. Ground 3 – Obviousness over Chee and Gahan

After considering all of the arguments and cited evidence of record, we determine that Petitioner has not demonstrated by a preponderance of the evidence that Chee and Gahan render claims 5, 6, and 8 obvious because Chee does not disclose the "collapsing step" recited in claim 1, which is a requirement of claims 5, 6, and 8 by virtue of their dependence on claim 1 as discussed above, and Petitioner does not further rely on Gahan as teaching the "collapsing step." Because Petitioner's obviousness ground relying on

Chee and Gahan fails for the above-described reasons, we need not address whether Petitioner shows that a person of ordinary skill in the art would have a reason to combine Chee and Gahan with a reasonable expectation of success. PO Resp. 49–51.

D. THE BIGNELL/MCCLOSKEY GROUNDS (GROUNDS 4–7)

Petitioner presents four grounds that rely on Bignell and McCloskey to teach the majority of the limitations of the asserted claims in Grounds 4 through 7. *See* Pet. 45–66. For Ground 4, Petitioner asserts that claims 1, 4–6, and 9 of the ’730 patent would have been obvious over Bignell and McCloskey. *Id.* at 45–60. For Ground 5, Petitioner asserts that claims 2 and 3 would have been obvious over Bignell, McCloskey, and Bentley. *Id.* at 61–63. For Ground 6, Petitioner asserts that claims 6 and 8 would have been obvious over Bignell, McCloskey, and Gahan. *Id.* at 63–65. For Ground 7, Petitioner asserts that claim 7 would have been obvious over Bignell, McCloskey, and Chiang. *Id.* at Pet. 65–66. For each of these grounds, Petitioner explains how each claim limitation is disclosed in the asserted references and how its assertions are supported by the testimony of Dr. Gabriel. *See id.* at 45–66; Ex. 1005 ¶¶ 233–344.

Patent Owner disputes Petitioner’s obviousness grounds relying on Bignell and McCloskey because the references do not teach the “correcting step” recited in claim 1, and Petitioner has not shown a person of ordinary skill in the art would have combined Bignell and McCloskey with a reasonable expectation of success. PO Resp. 52–61.

1. Ground 4 – Bignell and McCloskey

Petitioner asserts that claims 1, 4–6, and 9 of the ’730 patent would have been obvious over Bignell and McCloskey. Pet. 45–60. Petitioner

relies on Bignell as teaching methods for sequencing nucleic acid samples tagged with unique identifiers, and relies on McCloskey as teaching methods for correcting for errors or bias when amplifying nucleic acid samples. *Id.*

Patent Owner argues against Petitioner's Bignell and McCloskey grounds because the references do not disclose or teach the "collapsing step" recited in claim 1, and Petitioner does not show that a person of ordinary skill in the art would have combined Bignell and McCloskey. PO Resp. 52–61. We address each of Patent Owner's arguments below.

a) The "Collapsing Step" Recited in Claim 1

Petitioner relies on McCloskey as teaching the "collapsing step" recited in claim 1 of the '730 patent. Pet. 56–57. Petitioner contends McCloskey specifically teaches "correcting for errors or bias" using unique barcodes and data analysis. *Id.* at 56 (citing Ex. 1012 ¶ 2). Petitioner maps McCloskey's unique barcodes to the differentiator tags recited in claim 1. *Id.* at 56–57 (explaining that the '730 patent uses differentiator tags in the same way McCloskey uses unique barcodes). More specifically, Petitioner argues that McCloskey teaches the "unique barcodes allow it to 'correct[] for [the] skewing that can arise from PCR amplification[]' . . . by identifying duplicate barcodes in an amplified sample." *Id.* at 56 (citing Ex. 1012, ¶¶ 5, 53). Petitioner contends "[t]his allows McCloskey to determine that there are 'multiple amplified sequences derived from a single original molecule.'" *Id.* at 56–57 (citing Ex. 1012 ¶ 93).

According to Petitioner, "the '730 patent shows performing the 'collapsing' step by counting only unique tags to determine the number of times a sequence appeared in the original sample" such that "duplicates of a tag are identified but not included in the count." Pet. 57 (emphasis omitted, citing Ex. 1001, 30:48–51, 72:63–73:2). Based on this understanding,

Petitioner contends McCloskey teaches the “collapsing step” because “McCloskey’s barcodes allow it ‘to identify . . . redundant sequences and to quantify accurately the proportion of cells carrying a particular sequence variant by counting only distinctly tagged sequences.’” *Id.* (emphasis omitted, quoting Ex. 1012 ¶¶ 93, 106).

Patent Owner argues that Petitioner’s understanding of the “collapsing step” recited in claim 1 is incorrect and McCloskey does not teach the “collapsing step.” PO Resp. 52–56. Patent Owner contends the “collapsing step” requires collapsing combinations of targets and differentiator tags—not tags alone—because claim 1 recites “collapsing target:differentiator tag combinations.” *Id.* at 52 (citing Ex. 1001, 131:47–49). According to Patent Owner, the ’730 patent’s specification discloses that collapsing target:differentiator tag combinations provides advantages that counting tags alone does not, such as methods to make diploid genotype calls and more accurate consensus basecalls. *Id.* at 52–53 (citing Ex. 1001, 72:63–73:5, Fig. 5; Ex. 2023 ¶¶ 57, 182).

In light of this interpretation, Patent Owner argues McCloskey does not teach the “correcting step” because “McCloskey only discloses identifying tags alone without any analysis of the target to correct for errors or bias,” and “McCloskey does not disclose ‘collapsing’ reads together.” PO Resp. 53–54 (emphasis omitted). Patent Owner contends “McCloskey teaches detecting or ‘identifying’ a barcode to ‘count’ amplification products with distinct tags” but a person of ordinary skill in the art “would have understood that ‘identifying duplicate barcodes in an amplified sample’ and ‘counting only distinctly tagged sequences,’ is different and distinct from ‘collapsing target:differentiator tag combinations.’” *Id.* at 54–55 (citing Ex. 2023 ¶¶ 177–83).

Petitioner responds that the scope of the “collapsing step” includes collapsing on tags alone because claim 2 depends from claim 1 and recites “said correcting step further comprises collapsing all sequence reads comprising a same differentiator tag into a single read before determining a consensus base call.” Reply 4 (emphasis omitted). Petitioner argues that claim 1 must encompass collapsing using the differentiator tags alone because claim 2 depends from claim 1, and the specification includes an example of collapsing on tags alone in Figure 5. *Id.* at 4–5 (citing Ex. 1001, 73:3–5, Fig. 5).

Petitioner also asserts that McCloskey “collapses” reads because McCloskey teaches attaching unique barcodes to “uniquely identify a target nucleic acid in an experiment” and using the barcodes to identify duplicates and count the duplicates as one distinct read. Reply 5 (citing Ex. 1012 ¶ 21). Petitioner argues further that McCloskey collapses reads to remove the duplicates prior to counting, and “claim 1 does not require combining reads ‘together’; it only requires that duplicates be counted once (‘a single count’).” *Id.* at 7. Petitioner contends using target:tag combinations yields the same result as using tags alone because both processes collapse distinct target:tag combinations into a single count. *Id.* at 5–6.

Regarding Patent Owner’s argument that collapsing target:differentiator tag combinations provides the ability to make diploid genotype calls, Petitioner responds that the argument is irrelevant to claim 1 because claim 1 does not require making genotype calls. Reply 6. Petitioner contends “[e]ven if the ability to make genotype calls were required (it is not), McCloskey discloses its barcodes can improve making genotype calls on the FMR1 locus” because “McCloskey describes an example of testing the FMR1 locus, and explains that barcodes allow it ‘to quantify accurately

the proportion of cells carrying a particular sequence variant by counting only distinctly tagged sequences.” *Id.* at 6–7.

Patent Owner responds that McCloskey does not teach the “collapsing step” because “McCloskey counts only its unique barcodes to authenticate sequences, rather than collapsing barcode:sequence combinations to correct for errors.” Sur-Reply 23 (emphasis omitted). Patent Owner argues that “the plain language of Claim 1 requires ‘target:differentiator tag combinations,’ not tags alone.” *Id.* at 24. Patent Owner asserts that

collapsing “target:differentiator tag combinations” instead of barcodes alone—provides advantages when later making genotype calls . . . that highlight the difference between the claimed methods and the prior art. McCloskey does not recite making genotype calls, as Dr. Gabriel agrees (EX2025, 79:7-14), because that is not the purpose of its barcode-alone method for authenticating sequences, and therefore its method neither discloses collapsing “target:differentiator tag combinations” nor provides the advantages of the inventive method of the ’730 Patent.

Id. at 27 (emphasis omitted).

Regarding Petitioner’s argument that using target:tag combinations yields the same result as using tags alone, Patent Owner contends “Dr. Gabriel admitted that this is not true.” Sur-Reply 24. Patent Owner explains that

[w]hen Dr. Gabriel was presented with information that included both target sequences and barcodes . . . Dr. Gabriel was able to identify product sequences subject to amplification errors. . . . But when Dr. Gabriel was presented with information that included barcodes alone—like in McCloskey’s method—she was unable to do so.

Sur-Reply 24–26 (citing Ex. 2025, 46:22–50:8, 53:14–54:21; Ex. 2026; Ex. 2027).

We have considered both parties’ arguments and evidence and determine that Patent Owner is correct based on the evidence of record. As we have previously stated in the Chee grounds discussion, the plain language of claim 1 requires “collapsing target:differentiator tag combinations,” rather than collapsing only differentiator tags. The specification of the ’730 patent supports this interpretation. Figure 5 of the ’730 patent set forth below shows a method for genotyping based on target and 5 differentiator tag sequences. Ex. 1001, 19:35–36.

FIG. 5

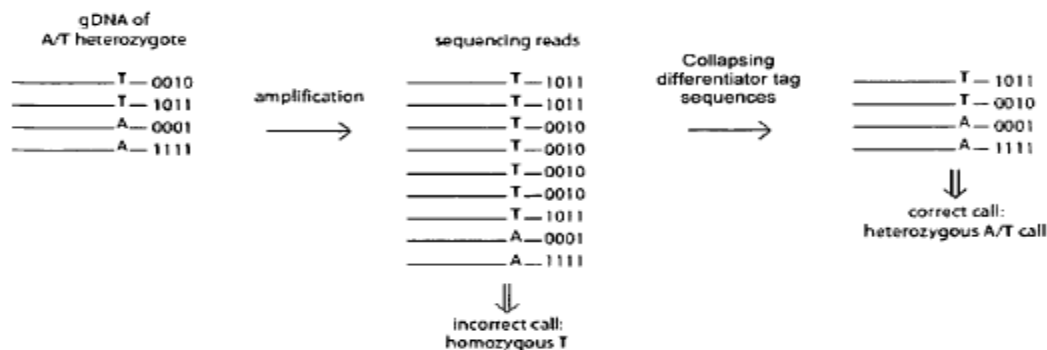


Figure 5 “depicts a method for making diploid genotype calls in which repeat target:differentiator tag combination are collapsed.” Ex. 1001, 73:3–5. It appears from Figure 5, that both the target sequences and the unique differentiator tags shown as combinations of 1s and 0s are used to determine which are repeat target:differentiator tag combinations that need to be collapsed to a single count.

By the same token, Example 2, titled “Use of Differentiator Tag Sequences to Detect and Correct Bias in a MIP-Capture Reaction of a Set of Exon Targets,” again shows use of both the target DNA and the unique differentiator tags to determine the count. *See* Ex. 1001, 71:25–73:5. Example 2 shows the construct of a table of target:differentiator tag

abundances from the read data as shown below. *Id.* at 72:50–51. This table appears below.

	Target ID	Differentiator Tag sequence ID	Count
1	1	3547	1
	2	4762	1
	1	9637	1
	1	1078	5
	3	4762	1
	1	2984	1

Ex. 1001, 72:53–62.

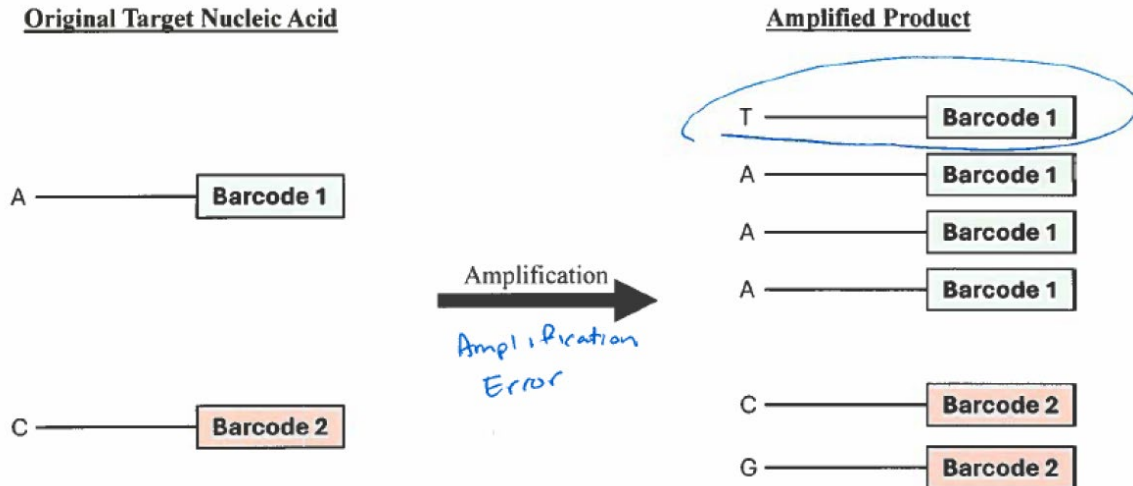
In analyzing the table shown above, the '730 patent states:

All 'count' entries should be '1', since any particular target-differentiator tag mapping will not occur more than once by chance, and therefore will only be observed if bias was present somewhere in the sample preparation process. For any target:differentiator tag combination observed more than once, all such reads are 'collapsed' into a single read before consensus basecalls are determined. This will cancel the effect of bias on consensus basecall accuracy.

Ex. 1001, 72:63–73:3.

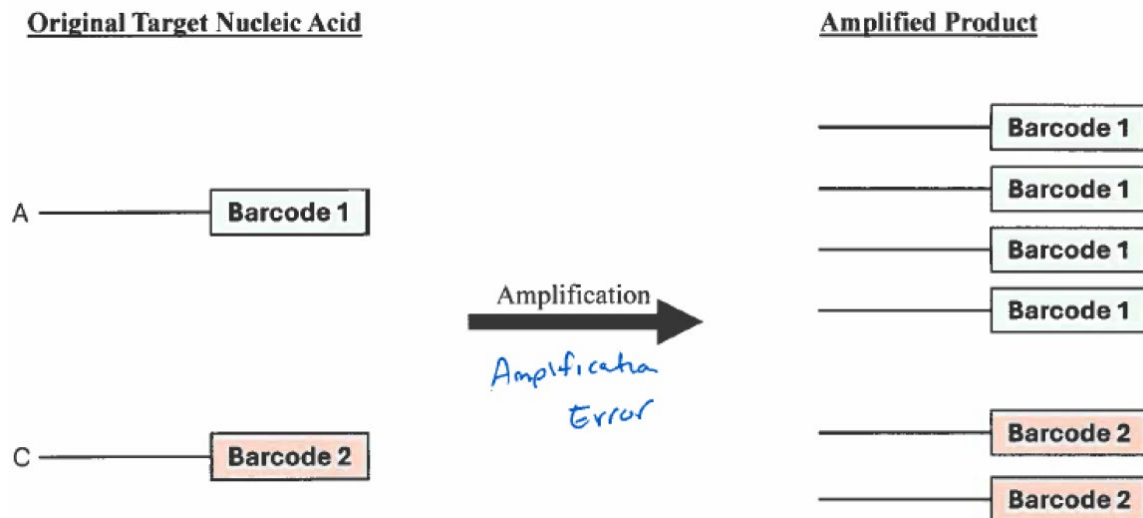
There are duplicate target IDs in the table that are included in individual counts because the differentiator tag sequence ID is different for these duplicates. The table shows that the differentiator tag sequence ID, not the tag alone, is used to determine the count.

Moreover, Patent Owner's example presented to Dr. Gabriel during her deposition underscores the difference between collapsing target:differentiator tag combinations and collapsing only differentiator tags. *Compare* Ex. 2026, *with* Ex. 2027. Exhibit 2026 is reproduced below.



Ex. 2026. Exhibit 2026 depicts two target nucleic acids with unique differentiator tags. The target nucleic acids undergo amplification and result in amplified products. Those products are reported as target:differentiator tag combinations.

Exhibit 2027 is reproduced below.



Ex. 2027. Exhibit 2027 depicts two target nucleic acids with unique differentiator tags. The target nucleic acids undergo amplification and result in amplified products. Only the differentiator tags of those products are reported. As shown in Exhibit 2026, if the products were collapsed on both targets and barcodes, then the target nucleic acid would be reported as both

T and A. As shown in Exhibit 2027, if the products were collapsed only on barcodes, then the target nucleic acid would be reported as either T or A, depending on which products were collapsed. As a result, we disagree with Petitioner's argument that using target:tag combinations yields the same result as using tags alone.

Dr. Gabriel confirms our conclusion. Dr. Gabriel testifies that you need not only the barcode, but also the sequence (as shown in Ex. 2026) to discover if there is an error. Dr. Gabriel testifies as follows with reference to Exhibit 2027.

Q. When the information provided to you is just the barcode alone, you cannot identify which sequence has an error resulting from an incorrectly incorporated nucleotide during amplification?

A. That's not right. You never just are analyzing – well, there's certain types of experiments called “counting applications” where you can analyze the barcorde to understand how many molecules are there. But that's aside. That's not what we're talking about here.

The things we're talking about here, you're using the barcode to make the bins to make – to line up sequences that have the same barcode. And then you can look at those sequences – they're not black lines, they're sequences – to find out if there's an error.

Q. In the scenario where you would be using just the barcode alone, such as a counting application – is that what you said?

A. Uh-hmm.

Q. So in the scenario in which you're using just the barcode alone, like in a counting application, you would not be able to identify error. Correct?

A. You wouldn't be trying to. It's a different type of experiment. That's a different application. Yeah.

Ex. 2025, 53:14–54:21 (objections omitted). In this colloquy, Dr. Gabriel confirms that to determine a sequence error, you cannot just use the barcode, but need the target sequence as well.

Dr. Puglisi agrees, and we credit his testimony as consistent with the record and the teachings of the '730 patent. Dr. Puglisi testifies:

A POSA would have understood that collapsing target:differentiator combinations to correct for error or bias is different than authenticating an amplification product by detecting a barcode sequence. For example, figure 5 depicts a method for making diploid genotype calls in which repeat target:differentiator tag combination[s] are collapsed” and the “effect of bias on consensus basecall accuracy” is canceled. EX1001, 73:2-5, Figure 5; *see also* EX1005, (Gabriel Decl.) ¶ 295 (citing EX1001, 28:32-44 (“[A] threshold number of independently isolated molecules (e.g., unique *combinations of differentiator tag and target sequences*) is analyzed to determine the genotype of a subject.”), 30:48-52 (“FIG. 5 depicts non-limiting method for genotyping based on *target and differentiator tag sequences*. Sequencing reads of *target and differentiator tags sequences* are collapsed to make diploid genotype calls.”), 72:63-73:2 (“For *any target:differentiator tag combination* observed more than once, *all such reads are ‘collapsed’ into a single read* before consensus basecalls are determined.”)). Unlike the methods to authenticate an amplification product in McCloskey, the '730 patent explains that using the sequence of the target nucleic acid molecule and a differentiator tag to collapse target:differentiator tag combinations provide methods to accurately make genotype calls and subsequently increase the accuracy when determining consensus base calls. EX 1001, 28:32-44, 30:48-52, 72:63-73:5, Figure 5. A POSA would have understood that McCloskey’s authenticating step does not use target nucleic acids and thus does not provide methods to make genotype calls or determine consensus base calls. Without the target nucleic molecule sequence, a POSA would not be able to make genotype calls or determine consensus base calls. EX 1001, 28:32-44, 30:48-52, 72:63-73:5, Figure 5.

Ex. 2023 ¶ 182.

Given this claim interpretation requiring collapsing target:differentiator tag combinations, it is undisputed that McCloskey, who uses unique barcodes alone, does not teach this limitation. Petitioner does not rely on Bignell as teaching this collapsing step. Accordingly, we conclude Petitioner has not shown that the combination of Bignell and McCloskey teaches the “collapsing step” recited in claim 1. Because claims 4 through 6 depend from claim 1, they also require the “collapsing step” of claim 1 that is not taught by the combination of Bignell and McCloskey.

b) The “Correcting Step” Recited in Claim 9

For independent claim 9, Petitioner states: “Most of the steps of claim 9 are identical to elements of claim 1, which Bignell and McCloskey disclose for the same reasons as claim 1. For the “*introducing* . . . “ and “*correcting* . . . “ steps, claim 9 uses slightly different words than claim 1.” Pet. 58–59. The correcting step of claim 9 requires “correcting for error or bias introduced during said workflow by recognizing target/differentiator tags having the same sequence as being derived from the same input molecule.” Ex. 1001, 132:56–59.

Petitioner relies on Bignell and McCloskey to teach claim 9’s requirement of “‘correcting for error or bias’ for the same reasons as claim 1.” Pet. 60. Petitioner states:

McCloskey identifies “redundant sequences” where multiple nucleic acids have the same barcode. EX1012(McCloskey), [0093], [0053]. It explains that the “ability to distinguish redundant from valid data is an immensely powerful tool.” *Id.*, [0053]. McCloskey determines that amplified nucleic acids that have the same barcode are “derived from a single original molecule.” *Id.*, [0093]; *id.*, [0106] (noting it can detect when “multiple amplified sequences are derived from a single

original molecule”). Thus, McCloskey teaches “*recognizing*” nucleic acids with the same unique barcode “*as being derived from the same input molecule.*” EX1005, ¶¶297-99.

Id.

Similarly, as we found with regard to our discussion of claim 1 where we found that McCloskey, who teaches using barcodes alone, does not teach the collapsing step, we also find here that McCloskey does not teach claim 9’s “correcting step” requiring “recognizing target/differentiator tags having the same sequence.” Ex. 1001, 132:56–59.

c) Conclusion Regarding Obviousness over Bignell and McCloskey

Based on the foregoing, we determine Petitioner has not shown by a preponderance of the evidence claims 1, 4–6, or 9 of the ’730 patent would have been obvious over Bignell and McCloskey. We find that Bignell and McCloskey do not teach claim 1’s “collapsing step,” which is also required by dependent claims 4–6, or claim 9’s “correcting step.”

2. Ground 5 – Obviousness over Bignell, McCloskey, and Bentley

Petitioner asserts that claims 2 and 3 of the ’730 patent would have been obvious over Bignell, McCloskey, and Bentley. Pet. 61–63. Claims 2 and 3 depend from claim 1. Ex. 1001, 131:28–57.

Patent Owner argues that for the same reasons discussed above for the combination of Bignell and McCloskey, “Bignell, McCloskey, and Bentley do not disclose claims 2 and 3 because the combination does not disclose claim 1,” and “Petitioner does not argue that the combination of Bignell, McCloskey, and Bentley provides any additional support for disclosing the elements of claim 1.” PO Resp. 61.

Petitioner again relies on McCloskey for teaching the required “collapsing step.” *See* Pet. 61–63. Because we have found that McCloskey does not disclose the “collapsing step” of claim 1, we determine that Petitioner has not demonstrated by a preponderance of the evidence that Bignell, McCloskey, and Bentley render claims 2 and 3 obvious because these claims depend from claim 1 and also require the missing “collapsing step.”

3. *Ground 6 – Obviousness over Bignell, McCloskey, and Gahan*

Petitioner asserts that claims 6 and 8 of the ’730 patent would have been obvious over Bignell, McCloskey, and Gahan. Pet. 63–65. Claims 6 and 8 depend from claim 1. Ex. 1001, 132:30–32, 132:36–38.

Patent Owner argues that for the same reasons discussed above for the combination of Bignell and McCloskey, “Bignell, McCloskey, and Gahan do not disclose claims 6 and 8 because the combination does not disclose claim 1,” and “Petitioner does not argue that the combination of Bignell, McCloskey, and Gahan provides any additional support for disclosing the elements of claim 1.” PO Resp. 62.

Petitioner again relies on McCloskey for teaching the required “collapsing step.” *See* Pet. 63–65. Because we have found that McCloskey does not disclose the “collapsing step” of claim 1, we determine that the Petitioner has not demonstrated by a preponderance of the evidence that Bignell, McCloskey, and Gahan render claims 6 and 8 obvious because these claims depend from claim 1 and also require the missing “collapsing step.”

4. *Ground 7 – Obviousness over Bignell, McCloskey, and Chiang*

Petitioner asserts that claim 7 of the '730 patent would have been obvious over Bignell, McCloskey, and Chiang. Pet. 65–66. Claim 7 depends from claim 1. Ex. 1001, 132:33–35.

Patent Owner argues that for the same reasons discussed above for the combination of Bignell and McCloskey, “the combination of Bignell, McCloskey, and Chiang does not disclose claim 7 because the combination does not disclose claim 1,” and “Petitioner does not argue that the combination of Bignell, McCloskey, and Chiang provides any additional support for disclosing the elements of claim 1.” PO Resp. 62.

Petitioner again relies on McCloskey for teaching the required “collapsing step.” *See* Pet. 65–66. Because we have found that McCloskey does not disclose the “collapsing step” of claim 1, we determine that the Petitioner has not demonstrated by a preponderance of the evidence that Bignell, McCloskey, and Chiang render claim 7 obvious because this claim depends from claim 1 and also requires the missing “collapsing step.”

III. CONCLUSION

For the foregoing reasons, we conclude Petitioner has not demonstrated by a preponderance of the evidence that claims 1–9 of the '730 patent are unpatentable.

IV. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that claims 1–9 of the '730 patent have not been shown to be unpatentable; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to this proceeding seeking judicial review of our decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

In summary:

Claim(s)	35 U.S.C. §	Reference(s)/ Basis	Claim(s) Shown Unpatentable	Claim(s) Not Shown Unpatentable
1–4, 9	102	Chee		1–4, 9
1–4, 7, 9	103	Chee		1–4, 7, 9
5, 6, 8	103	Chee, Gahan		5, 6, 8
1, 4–6, 9	103	Bignell, McCloskey		1, 4–6, 9
2, 3	103	Bignell, McCloskey, Bentley		2, 3
6, 8	103	Bignell, McCloskey, Gahan		6, 8
7	103	Bignell, McCloskey, Chiang		7
Overall Outcome				1–9

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